

Practice Problem Set

Introduction to Statistical Methods (ISM)

Instructions:

- Attempt all questions step-by-step.
 - For Naïve Bayes, show all vocabulary and probability calculations.
 - Use continuity corrections where applicable for Normal approximations.
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Module 1: Descriptive Statistics

Q1. (Data Interpretation & Outliers) Consider the following sample of processing times (in milliseconds) for a specific API call:

12, 15, 18, 14, 15, 12, 85, 16, 15, 19, 17

- Determine the five-number summary: Minimum, Q_1 , Median (Q_2), Q_3 , and Maximum.
- Determine potential outliers using the $1.5 \times IQR$ rule.
- Calculate the Mean and Standard Deviation.
- Based on the Mean and Median, interpret the skewness of the data.

Solution:

- Sort the Data:** 12, 12, 14, 15, 15, 15, 16, 17, 18, 19, 85. ($n = 11$)
 - **Minimum:** 12
 - **Median (Q_2):** The $(n + 1)/2 = 6^{th}$ value is **15**.
 - **First Quartile (Q_1):** The median of the lower half (12, 12, 14, 15, 15). The middle value is **14**.
 - **Third Quartile (Q_3):** The median of the upper half (16, 17, 18, 19, 85). The middle value is **18**.
 - **Maximum:** 85
- Outlier Check:**
 - $IQR = Q_3 - Q_1 = 18 - 14 = 4$.
 - Lower Bound = $Q_1 - 1.5 \times IQR = 14 - 1.5(4) = 14 - 6 = 8$.
 - Upper Bound = $Q_3 + 1.5 \times IQR = 18 + 1.5(4) = 18 + 6 = 24$.
 - Any value < 8 or > 24 is an outlier.

- **Conclusion:** The value **85** is an outlier.

(c) **Mean and Standard Deviation:**

- Mean $\bar{x} = \frac{\sum x}{n} = \frac{238}{11} \approx \mathbf{21.64}$.
- Variance $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$.

$$\sum x^2 = 12^2 + \dots + 85^2 = 9834$$

$$s^2 = \frac{9834 - 11(21.64)^2}{10} \approx \frac{9834 - 5151.4}{10} = \frac{4682.6}{10} = 468.26$$

$$s = \sqrt{468.26} \approx \mathbf{21.64}$$

(d) **Skewness:**

- Mean (21.64) > Median (15).
- **Interpretation:** The data is **positively skewed** (right-skewed), heavily influenced by the outlier (85).

Module 2: Probability Theory

Q2. (Conditional Probability & Bayes' Theorem) In a tech company, 30% of employees use Editor A, 50% use Editor B, and 20% use Editor C. The probability that an employee commits a syntax error given they use Editor A is 0.05. For Editor B, it is 0.03, and for Editor C, it is 0.08.

- Find the total probability that a randomly selected employee commits a syntax error.
- If an employee is found to have committed a syntax error, what is the probability they use Editor C?

Solution: Let E be the event of a syntax error. Let A, B, C be the events of using the respective editors. Given: $P(A) = 0.30, P(B) = 0.50, P(C) = 0.20$ $P(E|A) = 0.05, P(E|B) = 0.03, P(E|C) = 0.08$

(a) **Total Probability $P(E)$:**

$$P(E) = P(E|A)P(A) + P(E|B)P(B) + P(E|C)P(C)$$

$$P(E) = (0.05)(0.30) + (0.03)(0.50) + (0.08)(0.20)$$

$$P(E) = 0.015 + 0.015 + 0.016 = \mathbf{0.046}$$

(b) **Bayes' Theorem $P(C|E)$:**

$$P(C|E) = \frac{P(E|C)P(C)}{P(E)}$$

$$P(C|E) = \frac{(0.08)(0.20)}{0.046} = \frac{0.016}{0.046} \approx \mathbf{0.3478}$$

Module 3: Random Variables

Q3. (Piecewise Continuous Random Variable) A random variable X represents the delay in a system (in seconds) and has the following PDF:

$$f(x) = \begin{cases} kx & 0 \leq x \leq 2 \\ k(4-x) & 2 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of the constant k .
- (b) Find the probability $P(1 < X < 3)$.
- (c) Calculate the expected value $E[X]$.

Solution:

- (a) **Find k :** Area under PDF must be 1.

$$\int_0^2 kx \, dx + \int_2^4 k(4-x) \, dx = 1$$

$$k \left[\frac{x^2}{2} \right]_0^2 + k \left[4x - \frac{x^2}{2} \right]_2^4 = 1$$

$$k(2) + k[(16-8) - (8-2)] = 1$$

$$2k + k(8-6) = 2k + 2k = 4k = 1 \implies k = 0.25$$

- (b) **Find $P(1 < X < 3)$:**

$$P(1 < X < 3) = \int_1^2 0.25x \, dx + \int_2^3 0.25(4-x) \, dx$$

$$= 0.25 \left[\frac{x^2}{2} \right]_1^2 + 0.25 \left[4x - \frac{x^2}{2} \right]_2^3$$

$$= 0.25(2 - 0.5) + 0.25[(12 - 4.5) - (8 - 2)]$$

$$= 0.25(1.5) + 0.25(7.5 - 6) = 0.375 + 0.375 = 0.75$$

- (c) **Find $E[X]$:** Due to symmetry around $x = 2$, the mean is 2. Let's verify:

$$E[X] = \int_0^2 x(0.25x) \, dx + \int_2^4 x(0.25(4-x)) \, dx = 2$$

Q4. (Discrete Random Variable & Variance) Let X be a discrete random variable with PMF:

X	0	1	2	3
$P(X)$	0.1	$2c$	0.3	c

- (a) Find c .
- (b) Calculate $E[X]$ and $Var(X)$.

(c) Calculate $E[2X - 3]$.

Solution:

(a) $\sum P(X) = 1 \implies 0.1 + 2c + 0.3 + c = 1 \implies 3c = 0.6 \implies c = 0.2$.
The probs are: $P(0) = 0.1, P(1) = 0.4, P(2) = 0.3, P(3) = 0.2$.

(b) **Expectation** $E[X]$:

$$E[X] = 0(0.1) + 1(0.4) + 2(0.3) + 3(0.2) = 0 + 0.4 + 0.6 + 0.6 = \mathbf{1.6}$$

Variance $Var(X) = E[X^2] - (E[X])^2$:

$$E[X^2] = 0^2(0.1) + 1^2(0.4) + 2^2(0.3) + 3^2(0.2) = 0 + 0.4 + 1.2 + 1.8 = 3.4$$

$$Var(X) = 3.4 - (1.6)^2 = 3.4 - 2.56 = \mathbf{0.84}$$

(c) **Linear Combination:**

$$E[2X - 3] = 2E[X] - 3 = 2(1.6) - 3 = 3.2 - 3 = \mathbf{0.2}$$

Module 4: Sampling & Estimation

Q5. (Sample Size Estimation) A quality control engineer wants to estimate the mean life of a new battery. Preliminary data suggests the standard deviation is $\sigma = 15$ hours. How large a sample size is needed to estimate the mean with a margin of error of 2.5 hours at a 99% confidence level? (Use $Z_{0.005} = 2.576$)

Solution: Formula: $n = \left(\frac{Z \cdot \sigma}{ME}\right)^2$

- $Z = 2.576$
- $\sigma = 15$
- $ME = 2.5$

$$n = \left(\frac{2.576 \times 15}{2.5}\right)^2 = \left(\frac{38.64}{2.5}\right)^2 = (15.456)^2 \approx 238.88$$

Since sample size must be an integer, always round up:

$$\mathbf{n = 239}$$

Q6. (Sampling Distribution) The scores on a standardized test are normally distributed with $\mu = 75$ and $\sigma = 10$. If a random sample of 25 students is selected, what is the probability that the sample mean score is greater than 78?

Solution: We need $P(\bar{X} > 78)$.

- Standard Error $SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{25}} = 2$.
- Z-score transformation: $Z = \frac{\bar{X} - \mu}{SE} = \frac{78 - 75}{2} = \frac{3}{2} = 1.5$.

$$P(\bar{X} > 78) = P(Z > 1.5) = 1 - P(Z < 1.5)$$

From Z-table, $P(Z < 1.5) \approx 0.9332$.

$$P(Z > 1.5) = 1 - 0.9332 = \mathbf{0.0668}$$

Module 5: Naïve Bayes Classification

Q7. (Text Classification - Bag of Words) (*This question style appeared in the 2025 Regular Exam*)

You are given a training dataset of 4 short messages labeled as "Spam" or "Ham" (Not Spam).

ID	Message	Label
1	"Win cash now"	Spam
2	"Call me later"	Ham
3	"Win free cash"	Spam
4	"Meeting at five"	Ham

Use the Multinomial Naïve Bayes method to classify the new message: "**Win free call**". (Apply Laplace Smoothing if necessary, though for this calculation assume simple raw probabilities or use smoothing if counts are 0).

Solution:

Step 1: Calculate Priors

- Total Docs = 4. Spam = 2, Ham = 2.
- $P(\text{Spam}) = 2/4 = 0.5$
- $P(\text{Ham}) = 2/4 = 0.5$

Step 2: Build Vocabulary Calculate Likelihoods Vocabulary $V = \{\text{win, cash, now, call, me, later, free}\}$

Class Spam: Messages: "Win cash now", "Win free cash" Words: win(2), cash(2), now(1), free(1). Total words $N_{\text{Spam}} = 6$.

Class Ham: Messages: "Call me later", "Meeting at five" Words: call(1), me(1), later(1), meeting(1), at(1), five(1). Total words $N_{\text{Ham}} = 6$.

Step 3: Calculate Probabilities for New Message ("Win", "free", "call") Using Laplace Smoothing (+1 to count, +|V| to denominator) where $|V| = 10$. Denominator for Spam = $6 + 10 = 16$. Denominator for Ham = $6 + 10 = 16$.

Likelihood for Spam:

- $P(\text{win}|\text{Spam}) = (2 + 1)/16 = 3/16$
- $P(\text{free}|\text{Spam}) = (1 + 1)/16 = 2/16$
- $P(\text{call}|\text{Spam}) = (0 + 1)/16 = 1/16$
- $P(\text{Message}|\text{Spam}) \propto 0.5 \times \frac{3}{16} \times \frac{2}{16} \times \frac{1}{16} = 0.5 \times \frac{6}{4096} \approx \mathbf{0.000732}$

Likelihood for Ham:

- $P(\text{win}|\text{Ham}) = (0 + 1)/16 = 1/16$
- $P(\text{free}|\text{Ham}) = (0 + 1)/16 = 1/16$
- $P(\text{call}|\text{Ham}) = (1 + 1)/16 = 2/16$
- $P(\text{Message}|\text{Ham}) \propto 0.5 \times \frac{1}{16} \times \frac{1}{16} \times \frac{2}{16} = 0.5 \times \frac{2}{4096} \approx \mathbf{0.000244}$

Step 4: Conclusion Since $0.000732 > 0.000244$, the message is classified as **Spam**.